

Project ID:

J26_IT_302

1. Topic (12 words max)

AI-Driven IoT Smart Fisherman Safety System for Real-Time Emergency Detection

2. Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	J26_IT_302
Year	2026

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Fishing is one of the most important economic activities in Sri Lanka and serves as the primary source of income for thousands of families living in coastal communities. Despite its importance, fishing remains a high-risk occupation due to the unpredictable nature of the marine environment. Fishermen are frequently exposed to dangerous situations such as sudden weather changes, boat capsizing, engine failures, water leakage, onboard fires, and accidental falls overboard. These incidents often occur far from the shoreline, where access to immediate assistance is limited and communication networks are unreliable.

Currently, most small-scale fishing boats in Sri Lanka rely on mobile phones or conventional radio systems as their primary means of communication during emergencies. However, these methods depend on manual operation and stable network connectivity, which cannot always be guaranteed in offshore areas. As a result, emergency situations are often reported late, accurate boat location information may not be available, and rescue authorities face difficulties in responding quickly and efficiently. Furthermore, existing systems do not provide continuous monitoring of boat conditions or intelligent analysis to identify potential safety risks before they become critical.

This research belongs to the **Autonomous Intelligent Machine Systems** research cluster and focuses on the maritime safety and IoT industry. The proposed solution also contributes towards the United Nations Sustainable Development Goals, particularly **SDG 3 (Good Health and Well-being)** by improving emergency response, **SDG 9 (Industry, Innovation and Infrastructure)** through smart IoT technologies, and **SDG 14 (Life Below Water)** by supporting safer fishing activities.

6. Brief description of Existing Research and Systems (300 words max)

Several research studies have proposed IoT-based systems to improve the safety of fishermen by using GPS, wireless communication, and emergency alert mechanisms. The first IEEE study presents a smart fisherman safety system that combines IoT technologies to provide location tracking, emergency communication, and real-time monitoring. The system mainly focuses on improving communication between fishermen and rescue authorities during emergency situations using wireless technologies. It demonstrates how IoT can enhance maritime safety by providing timely alerts and location information. However, the system mainly concentrates on emergency communication and does not monitor the overall condition of the fishing boat or fish storage.

The second IEEE study introduces an IoT-based communication and protection system for fishermen using GPS and wireless communication technologies. The system helps fishermen share their location and send emergency alerts, improving rescue operations and reducing response time. Although it provides an affordable communication solution, it has limited monitoring capabilities and does not include intelligent analysis of multiple sensor data or advanced visualization features.

Most existing systems primarily focus on GPS tracking, border monitoring, or emergency communication. They generally lack comprehensive monitoring of boat health, including engine temperature, smoke, fire, vibration, battery condition, and water leakage. In addition, fish storage conditions such as temperature, humidity, door status, and storage level are rarely monitored.

To overcome these limitations, the proposed **SeaGuardian** system integrates IoT sensors, LoRa-based communication, boat health monitoring, fish storage monitoring, Digital Twin visualization, AI-based risk analysis, cloud services, and mobile applications into a single intelligent platform. This provides a more comprehensive, reliable, and affordable safety solution for small-scale fishermen.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The proposed solution, SeaGuardian, is an AI-driven IoT-based smart fisherman safety and emergency communication system designed to improve the safety of small-scale fishermen in Sri Lanka. The system integrates IoT sensors, LoRa-based communication, cloud computing, AI, a Digital Twin, and Android applications to provide real-time monitoring, intelligent risk assessment, and emergency response support.

The system consists of main modules. The first module is the SeaGuardian IoT Device, which is installed on the fishing boat. It uses an ESP32 microcontroller to collect data from multiple sensors, including a GPS module for live location tracking, an MPU6050 sensor for monitoring boat tilt and motion, a water sensor to detect water entering the boat, a DS18B20 temperature sensor, a battery monitoring circuit, and an SOS button for manual emergency alerts.

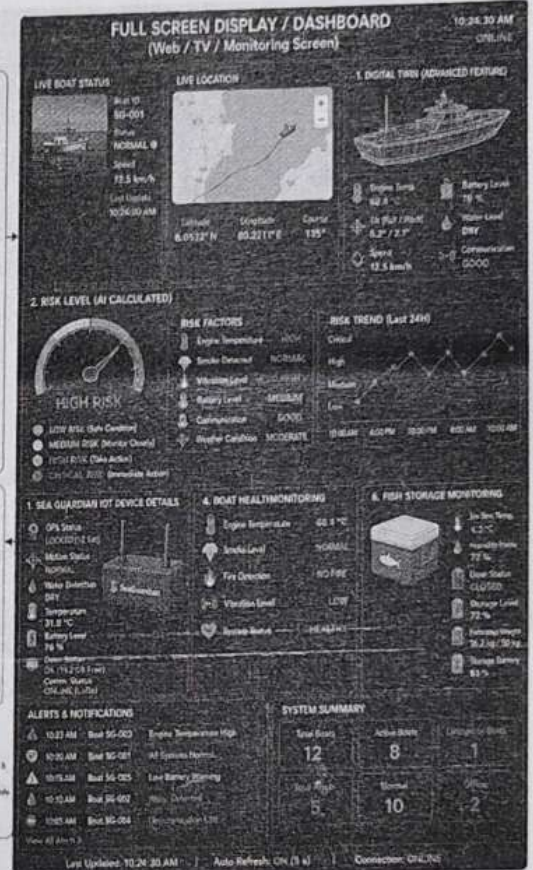
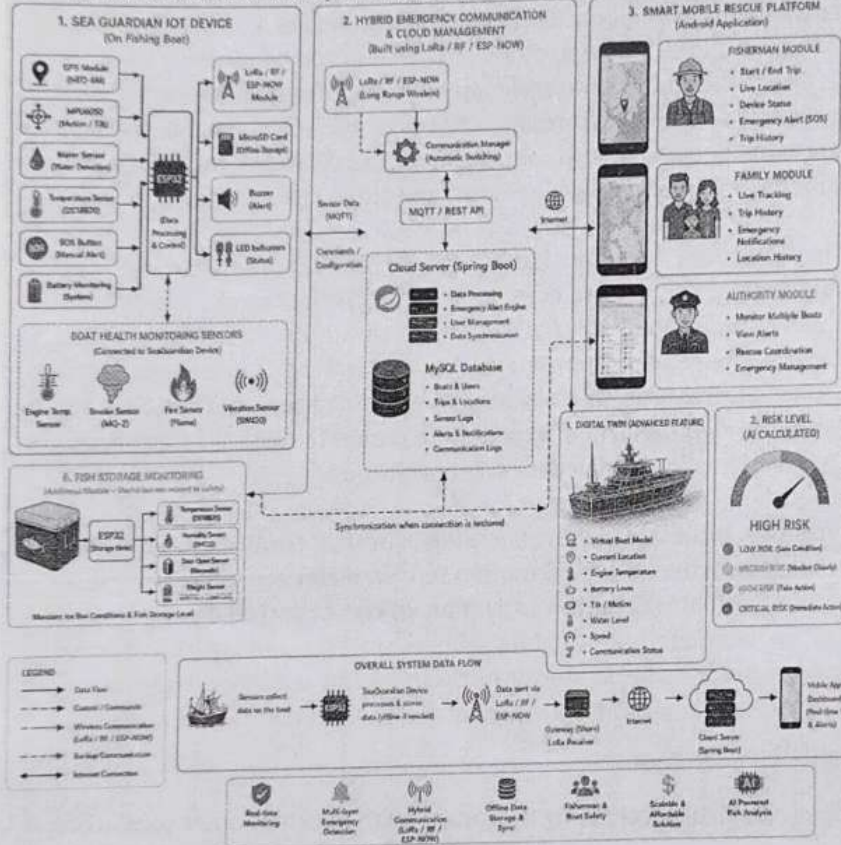
The second module is the Hybrid Emergency Communication System, which uses LoRa, RF, and ESP-NOW technologies to transmit data over long distances when mobile network coverage is unavailable. A Communication Manager automatically selects the most suitable communication method.

The third module is the Smart Mobile Rescue Platform, which provides separate Android interfaces for fishermen, family members, and rescue authorities. Fishermen can monitor their boat status, receive alerts, and send SOS messages. Family members can track the boat's location and receive emergency notifications, while authorities can monitor multiple boats and coordinate rescue operations.

The fourth module is the Boat Health Monitoring System, which continuously monitors engine temperature, smoke, fire, and vibration to identify mechanical problems before they become dangerous. The fifth module is the Fish Storage Monitoring System, which monitors ice box temperature, humidity, door status, storage level, and battery condition to help preserve fish quality and reduce financial losses.

SeaGuardian – Conceptual Diagram

AI-Driven IoT Smart Fisherman Safety and Emergency Communication System



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The successful implementation of the SeaGuardian project requires knowledge in IoT hardware development, embedded systems, mobile application development, backend software engineering, cloud communication, and system testing. The team members will gain the required technical knowledge through university modules, online learning platforms, official hardware documentation, and research publications related to IoT, GPS communication, mobile application development, and cloud computing.

The IoT hardware development requires knowledge of ESP32 programming, GPS integration, sensor interfacing, and 4G/LTE communication. The backend development requires experience with Spring Boot, REST APIs, MQTT communication, and MySQL database management. Mobile application development will be carried out using Android Studio, focusing on real-time GPS visualization, emergency notifications, and user management. System integration and testing require understanding of sensor fusion logic, communication reliability, and offline data synchronization.

The data required for this project will be generated through controlled laboratory simulations. Sensor values from the MPU6050 motion sensor, water sensor, temperature sensor, GPS module, and manual SOS button activation will be collected during testing. Network failures will be simulated to validate offline storage and synchronization functionality. The collected data will be stored in the backend database and used to evaluate response time, communication success rate, detection accuracy, and synchronization reliability.

Since the research does not involve collecting personal health information or conducting experiments on human participants, formal ethical clearance is not expected to be required. However, if user information such as location or personal details is collected during future deployment, appropriate privacy and data protection measures should be considered.

9. Objectives and Novelty

Main Objective To design and develop a low-cost IoT-based smart fisherman safety system with real-time tracking and simulated emergency detection to improve the safety of Sri Lankan fishermen and support faster emergency response.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT23257054 Kumaratunga R.C.S	Design and develop the SeaGuardian IoT Device for real-time onboard monitoring and edge processing.	• Select suitable IoT hardware components. • Design and develop the ESP32-based embedded system. • Integrate GPS (NEO-6M) for live location tracking. • Integrate MPU6050 to monitor boat tilt and motion. • Implement water leakage detection. • Implement temperature monitoring using DS18B20. • Develop battery monitoring system. • Implement SOS emergency button with buzzer and LED alerts. • Develop offline data storage using MicroSD. • Implement edge data processing and sensor filtering. • Design waterproof hardware enclosure. • Perform hardware calibration and testing. • Evaluate power consumption and system reliability/laboratory demonstrations.	The SeaGuardian IoT device integrates multiple sensors, including GPS, motion, water level, temperature, battery monitoring, and emergency controls, into a single intelligent embedded platform. Unlike existing systems that mainly focus on GPS tracking, this prototype performs edge computing by processing sensor data locally, allowing the system to continue monitoring even when internet connectivity is unavailable. The device also supports offline data storage, automatic synchronization, and real-time sensor fusion, making it more reliable for remote marine environments where communication is limited.

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V2.2

		• Test the functionality of every sensor before integrating with the communication module.	
IT23322912 Weerathunga V.K	Develop the hybrid communication infrastructure, cloud platform, Digital Twin, and AI-based risk analysis.	• Implement LoRa communication. • Develop RF and ESP-NOW communication modules. • Design automatic communication switching algorithm. • Develop Spring Boot backend APIs. • Design MySQL database. • Implement cloud data synchronization. • Develop real-time monitoring dashboard APIs. • Build the Digital Twin of the fishing boat. • Synchronize real sensor data with the Digital Twin. • Develop AI-based Risk Level Assessment. • Analyze multiple sensor values simultaneously. • Classify risk as Low, Medium, High, or Critical. • Generate intelligent alerts and recommendations. • Evaluate communication reliability and AI performance.	This module introduces a hybrid communication architecture that combines LoRa, RF, and ESP-NOW with an automatic communication management system capable of selecting the most reliable communication channel based on network availability. In addition, it introduces a Digital Twin that provides a real-time virtual representation of the fishing boat by synchronizing live sensor data with a cloud platform. An AI-based Risk Assessment Model continuously analyzes multiple sensor readings and predicts the overall safety condition as Low, Medium, High, or Critical Risk, enabling intelligent decision-making instead of simply displaying raw sensor values.

IT23153486 Navodyani W.M.B	Develop the Flutter-based rescue platform and intelligent fish storage monitoring system.	• Develop flutter application using modern UI/UX. • Implement Fisherman module. • Implement Family monitoring module. • Implement Authority monitoring module. • Integrate live GPS tracking. • Display real-time boat information. • Implement emergency notifications. • Implement trip history and reports. • Develop Fish Storage Monitoring module. • Monitor temperature and humidity. • Detect storage door status. • Calculate fish storage level using load cell. • Display fish quality indicators. • Generate storage alerts and reports. • Evaluate application usability and performance.	The mobile platform is designed for three different user groups—fishermen, family members, and rescue authorities—providing personalized interfaces and real-time information based on their roles. In addition to emergency monitoring, the system introduces smart fish storage monitoring, which continuously measures temperature, humidity, storage level, battery condition, and storage door status to maintain fish quality throughout the fishing journey. This combination of safety monitoring and post-harvest quality management is rarely found in existing fisherman safety solutions.
IT23182822 Wathsala K.L.K	Design and develop the Boat Health Monitoring System for predictive fault detection and early warning.	• Develop Boat Health Monitoring. • Integrate engine temperature sensor. • Integrate smoke sensor. • Integrate flame sensor. • Integrate vibration sensor. • Monitor engine operating conditions. • Detect abnormal vibration patterns. • Detect smoke and fire hazards.	This module extends traditional fisherman safety systems by introducing boat health monitoring and predictive maintenance. It continuously monitors engine temperature, smoke, fire, and vibration to identify abnormal operating conditions before they

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Topic Assessment Form

V2.2

		• Generate early warning alerts. • Send health status to the SeaGuardian • Produce maintenance reports. • Develop predictive maintenance logic. • Evaluate sensor accuracy and response • Test fault detection under simulated conditions. • Validate system reliability. • Prepare the complete testing and evaluation report.	become critical failures. Rather than generating alerts only after an emergency occurs, the system provides early warning notifications and maintenance recommendations based on sensor analysis, helping prevent accidents, reduce equipment damage, and improve the reliability of fishing boats.
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**10. This section is to be completed by the
Supervisor and the Co-supervisor of the Project.**

- a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	☒	No	☐
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- b) The proposed topic exhibits novelty.

Yes	☒	No	☐
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- c) The student group can successfully execute the proposed project.

Yes	☒	No	☐
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- d) The proposed sub-objectives reflect the students' areas of specialization.

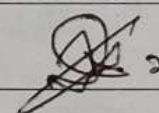
Yes	☐	No	☒
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- e) Data required for the project is available.

Yes	☐	No	☒
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- f) Any other comments:

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	Title	First Name	Last Name	Signature
Supervisor*	Prof.	Samarth	Rajesh	
Co-Supervisor*	Mr.	Nelum	Anura Sena	 20/07/2026
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

*- At this stage, identifying the Co-Supervisor is compulsory. Students should reach out to a staff member, with guidance from their supervisor, to serve as the co-supervisor for this project.

This part is to be filled by the Topic Screening Panel members.

Project ID:

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes (should be followed up by the supervisor)*	✓
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

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The Review Panel Details

Member's Name	Signature	Date
Saijeeri Chandrasiri	Saijeeri	27/07/2026
Tharushi Rubasinghe	Tharushi	27/07/2026

*Important:

1. According to the comments given by the panel, make the necessary modifications and get approval from the **Supervisor** or the **Same Panel**.
2. If the project topic is rejected, identify a new topic, and follow the same procedure until the topic is approved by the assessment panel.